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Interim Python Project Report

On

Language Detection Capstone Project

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Introduction:

The **Language Detection App** is a Python-based graphical user interface (GUI) application designed to detect the language of a given text input. This app utilizes machine learning to classify the language using a **Naive Bayes** model trained on a dataset of text samples in various languages. The application allows users to enter text, automatically detect its language, and access more information about the detected language via a Google search.

Objective and Scope:

The primary objective of the Language Detection App is to provide an intuitive and engaging tool that allows users to easily detect the language of a given text using a machine learning model. The app aims to accurately classify the language of input text through a pre-trained Naive Bayes model and offer users the ability to explore more information about the detected language by redirecting them to a Google search. Additionally, the app focuses on creating a user-friendly interface that simplifies navigation between different sections, such as the main menu, language detection, and the help/about page, ensuring a smooth and efficient user experience.

SCOPE:

The scope of the **Language Detection App** encompasses the detection of text language using a machine learning model trained on a diverse set of languages. It provides users with a simple and interactive interface to input text, view the detected language, and access more detailed information through Google search links. The app is designed to be extensible, with potential for supporting more languages, adding real-time translation features, and incorporating multi-language detection in the future. The app is built using **Python**, with **Tkinter** for the graphical user interface (GUI), and **Scikit-learn**, **Pandas**, and **NumPy** for machine learning and data processing. It caters to a broad audience, including language enthusiasts, educators, developers, and travelers

Application Tools:

The **Language Detection App** uses the following tools:

1. **Python:** The main programming language used for the app's development.
2. **Tkinter:** A GUI library in Python for creating interactive windows, buttons, and text input fields.
3. **Scikit-learn:** A machine learning library for training the **Naive Bayes** model and using **Count Vectorizer** for text classification.
4. **Pandas:** A data manipulation library for loading and processing the dataset.
5. **NumPy:** A library for handling arrays and numerical operations in machine learning.

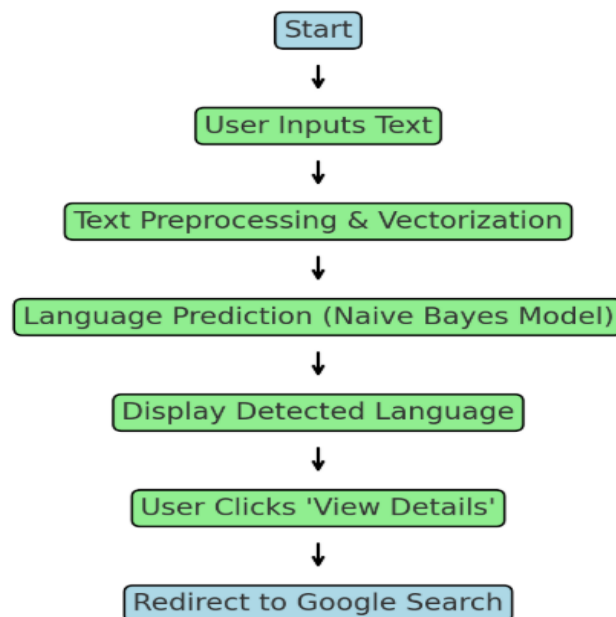
6. **Web Browser (web browser library):** Used to open Google search results for more information about the detected language.
7. **CSV File:** Stores the dataset with text samples and language labels used to train the model.

Flow Chart:

Here is a simple flowchart of the Stylish Language Detection App:

1. Start: The process begins.
2. User Inputs Text: The user enters text for language detection.
3. Text Preprocessing & Vectorization: The input text is processed and converted into a numerical format suitable for the machine learning model.
4. Language Prediction (Naive Bayes Model): The text is passed through the trained Naive Bayes model to predict the language.
5. Display Detected Language: The detected language is shown to the user.
6. User Clicks 'View Details': The user clicks on the "View Details" button to explore more information about the language.
7. Redirect to Google Search: The app redirects the user to a Google search for more details about the detected language.

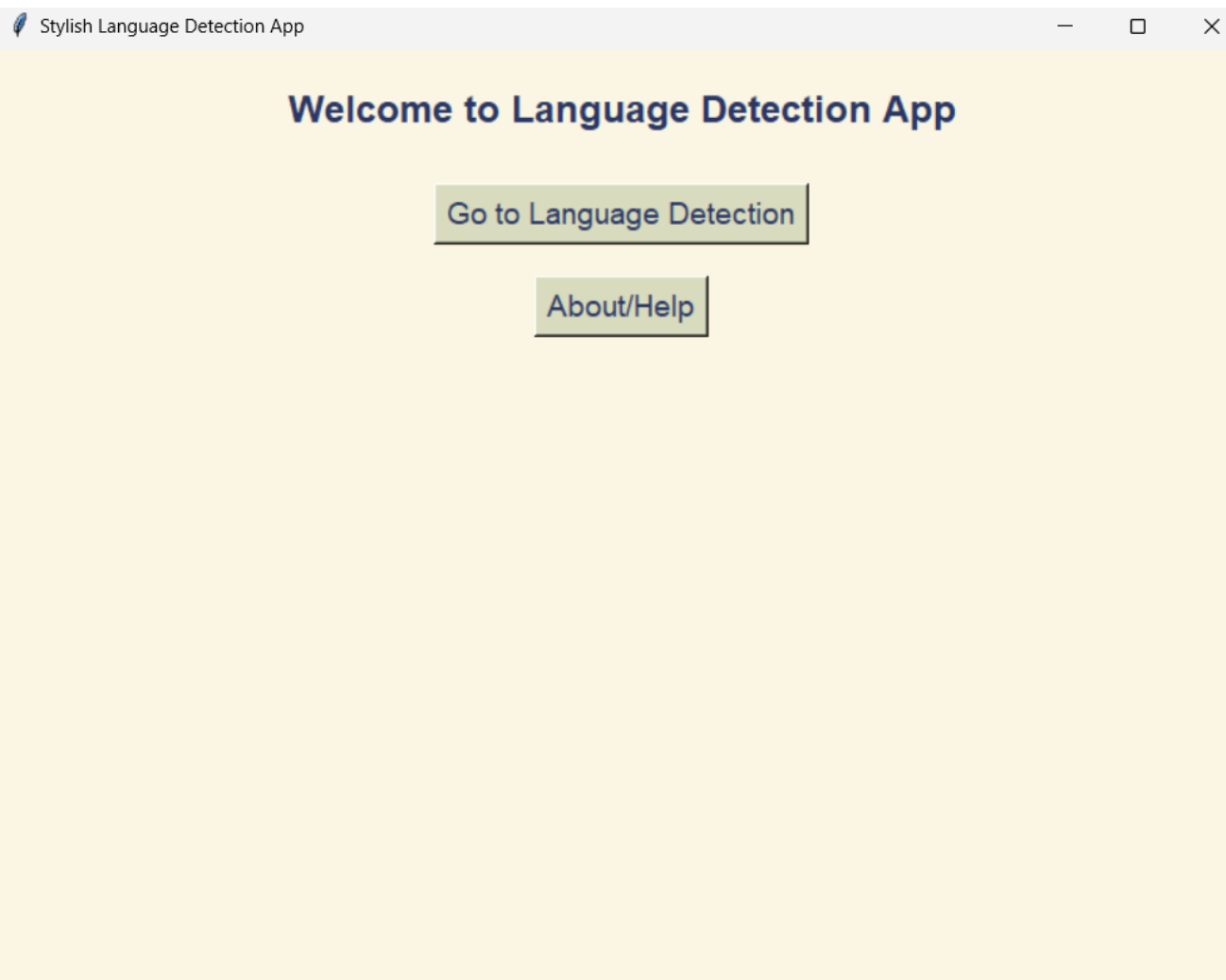
This flow ensures a seamless user experience with easy navigation and accurate language detection.



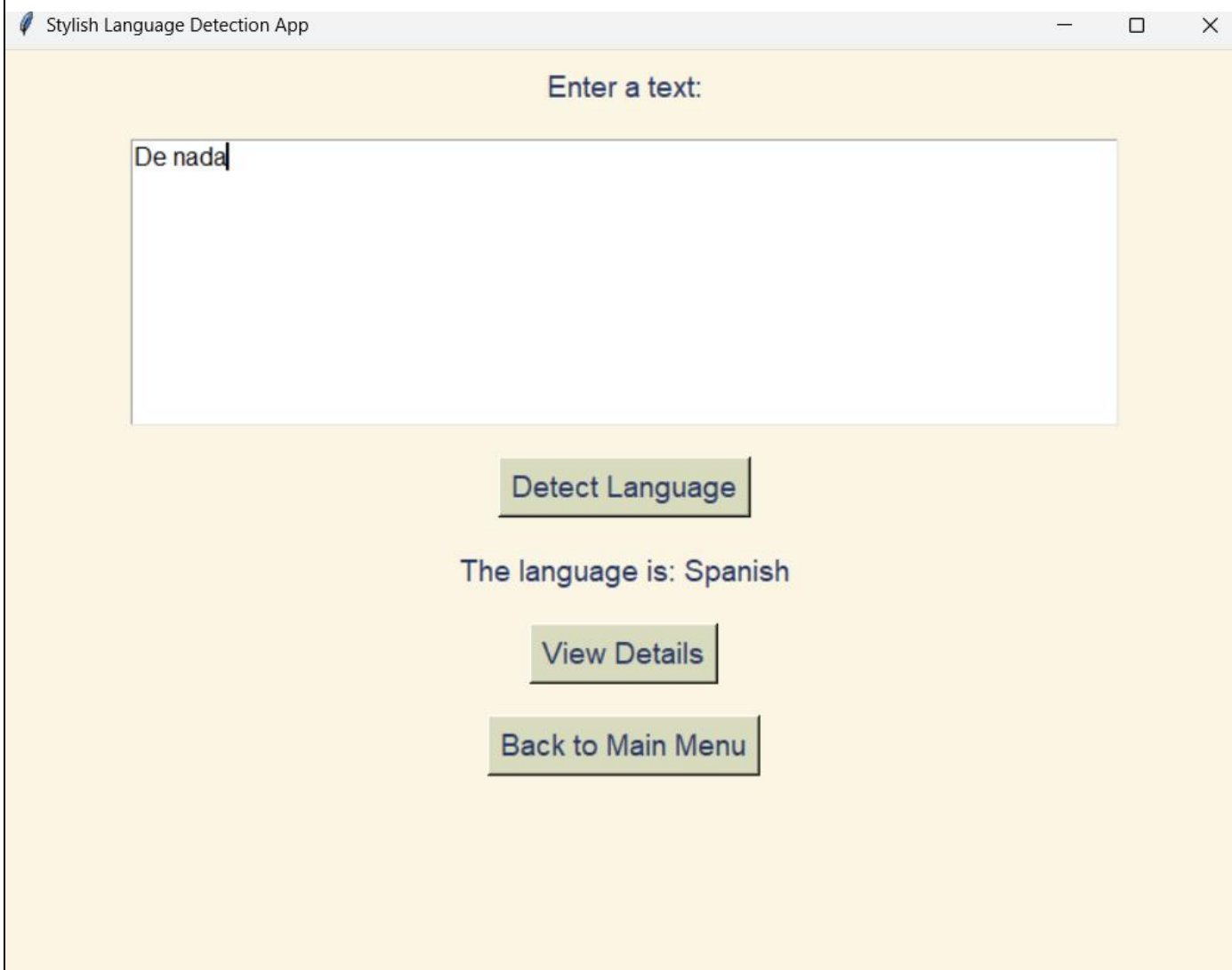
Preview:

The Language Detection Project is a tool designed to identify the language of a given text input with accuracy and efficiency. It supports multiple languages, making it suitable for diverse applications like multilingual content sorting and text preprocessing for Natural Language Processing (NLP) tasks. The system features a simple command-line interface where users can input text directly or analyse a text file, with results generated quickly thanks to its lightweight design. Built using Python, the project leverages libraries like `Lang detect` or `languid` for detection, alongside optional tools for text preprocessing and argument handling. This extendable framework allows for easy addition of new languages, making it versatile for personal or professional use. Its core functionality involves preprocessing the input, analyzing it through detection algorithms, and returning the detected language, often with confidence levels for enhanced reliability.

Starting menu interface:



Main Interface:



The screenshot shows a web application window titled "Stylish Language Detection App". The interface has a light yellow background. At the top, it says "Enter a text:". Below this is a large white text input box containing the text "De nada". Under the input box is a green button labeled "Detect Language". Below the button, the text "The language is: Spanish" is displayed. At the bottom, there are two more green buttons: "View Details" and "Back to Main Menu".

Conclusion:

The **Language Detection Project** offers a practical and efficient solution for identifying the language of textual data. With its lightweight design, multi-language support, and user-friendly command-line interface, it is well-suited for a variety of applications, from NLP preprocessing to multilingual content management. Its extensibility ensures that it can grow to support additional languages, making it a versatile tool for both personal and professional use. This project serves as a robust foundation for language-based text analytics, providing accurate and reliable results in a straightforward and accessible manner.